



Microcontroller System Specification Report

PIC16F877A Microcontroller

Report Topics

PIC16F877A Pin Description
CPU Architecture and Main Blocks
RA4 LED Interface Troubleshooting
PIC16F877A vs. ATmega328P Comparison

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Training

Embedded Systems
PIC16F877A

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1 Introduction

Microcontrollers are one of the fundamental building blocks of modern embedded systems. Unlike general-purpose computers, microcontrollers integrate a processor, memory, input/output interfaces, timers and communication peripherals into a single integrated circuit.

The selection of a suitable microcontroller is therefore an important stage in the system specification phase of an embedded project.

For this project, the selected device for study is the **PIC16F877A**, an 8-bit microcontroller manufactured by Microchip Technology.

The PIC16F877A is widely used in educational projects, industrial control applications and embedded-system prototypes because it provides a relatively large number of I/O pins, analog inputs, timers, serial communication peripherals and non-volatile memory.

The objectives of this report are:

- Describe all pins of the PIC16F877A.
- Explain the main blocks of its CPU architecture.
- Investigate why an LED connected to RA4 may not work correctly.
- Compare the PIC16F877A with the ATmega328P.
- Identify applications where the ATmega328P can be a better choice.

1.1 Main Specifications

Table 1: Main PIC16F877A specifications

Specification	Description
CPU	8-bit PIC RISC architecture
Instruction Set	35 single-word instructions
Maximum Clock	Up to 20 MHz
Program Memory	Up to 8K × 14-bit Flash words
RAM	368 bytes
EEPROM	256 bytes
ADC	10-bit, up to 8 channels
I/O	Up to 33 general-purpose I/O pins
Timers	Timer0, Timer1 and Timer2
CCP Modules	Two CCP modules
Serial Communication	USART, SPI and I ² C capability
Package	40-pin DIP and other package options

2 Part (a): PIC16F877A Pin Description

The PIC16F877A is commonly available in a 40-pin package. The pins provide power, reset, oscillator, digital I/O, analog inputs, communication interfaces and programming/debugging functions.

2.1 PIC16F877A Pinout

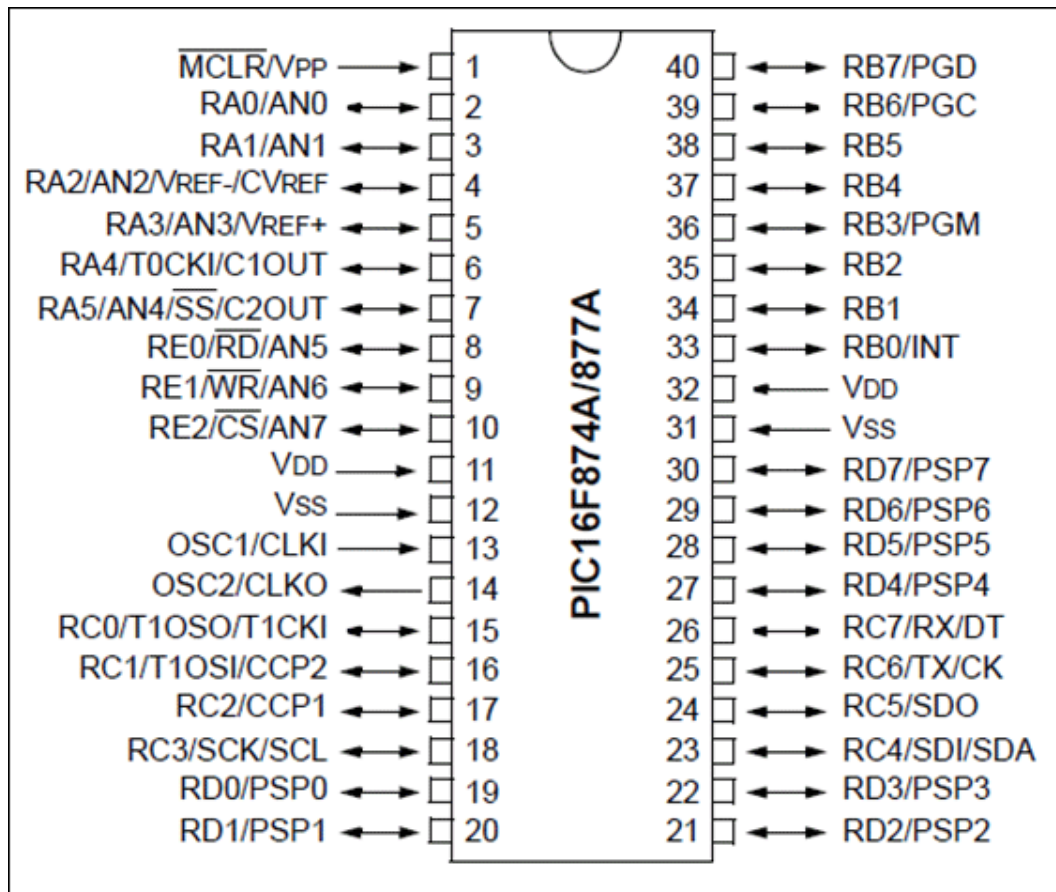


Figure 1: PIC16F877A 40-pin simplified pinout.

3 Part (b): Main CPU Blocks

The PIC16F877A contains several CPU blocks that cooperate to fetch, decode and execute instructions.

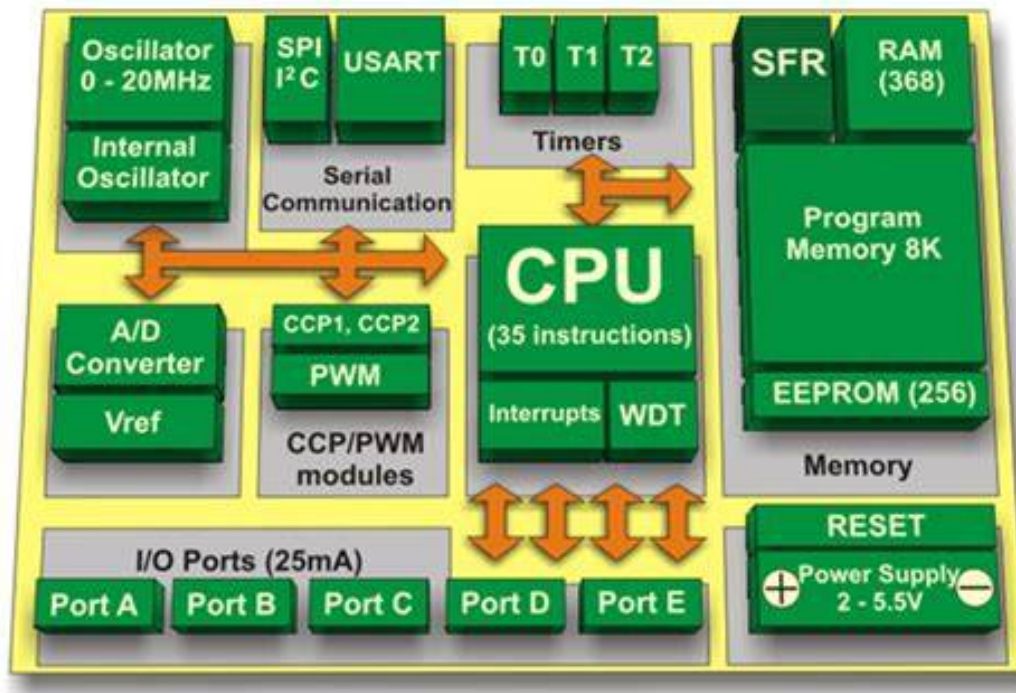


Figure 2: Simplified PIC16F877A CPU architecture.

3.1 Arithmetic Logic Unit (ALU)

The Arithmetic Logic Unit is responsible for performing arithmetic and logical operations. Typical operations include:

- Addition.
- Subtraction.
- Increment and decrement.
- AND operation.
- OR operation.
- XOR operation.
- Bit manipulation.
- Comparison-related operations.

The ALU operates mainly on 8-bit data.

The result can be stored in the W register or a file register, depending on the instruction.

The ALU also affects important bits in the STATUS register.

3.2 STATUS and Control

The STATUS register contains important information about the processor state.

Important flags include:

- **Z – Zero flag:** indicates that an operation produced a zero result.
- **C – Carry flag:** indicates a carry from the most significant bit during an arithmetic operation.
- **DC – Digit Carry:** indicates a carry from the lower four bits.

STATUS also contains bits related to memory-bank selection and processor operating conditions.

3.3 Program Counter

The Program Counter (PC) is a 13-bit register that contains the address of the next instruction to be executed.

The PIC16F877A can address up to 8K instruction words, therefore the 13-bit program counter is sufficient for addressing its program memory.

During normal sequential execution, the program counter is incremented automatically.

Instructions such as:

- GOTO
- CALL
- RETURN
- RETFIE
- Interrupt operations

can modify the normal program flow.

3.4 Flash Program Memory

Flash program memory stores the microcontroller firmware.

The PIC16F877A provides up to:

$$8K \times 14\text{-bit}$$

Flash program memory.

Flash memory is non-volatile, meaning that the program remains stored when the power supply is removed.

This makes it suitable for embedded applications where firmware must remain available after power cycling.

3.5 Instruction Register

The Instruction Register is associated with the currently fetched instruction.

The Program Counter identifies the address of the instruction in program memory.

The instruction is then fetched and made available to the CPU control logic for decoding and execution.

The instruction register therefore forms an important part of the connection between program memory and the execution hardware.

3.6 Instruction Decoder

The Instruction Decoder interprets the opcode contained in the instruction.

Its main tasks are:

- Identify the instruction type.
- Identify the operands.
- Generate internal control signals.
- Select the required ALU operation.
- Control data movement between registers.

The decoder allows the CPU to transform a binary instruction into the appropriate sequence of internal operations.

4 Part (c): LED Connected to RA4

One of the most common mistakes when using the PIC16F877A is assuming that every digital output pin behaves exactly the same.

RA4/T0CKI is a special case.

4.1 Why RA4 Can Cause Problems

RA4 has an **open-drain output structure**.

Unlike a conventional push-pull output, an open-drain output does not actively drive the output HIGH in the same way.

Therefore, if an LED is connected to RA4 using the same circuit normally used for another GPIO pin, the LED may:

- Remain permanently OFF.
- Remain permanently ON.
- Flash with reversed logic.
- Appear to flash incorrectly.

4.2 Correct RA4 LED Connection

A common method is to connect:

$$\text{VDD} \rightarrow \text{Resistor} \rightarrow \text{LED} \rightarrow \text{RA4}$$

When RA4 is driven LOW, it sinks current and the LED turns ON.

When RA4 is released, the current path is removed and the LED turns OFF.

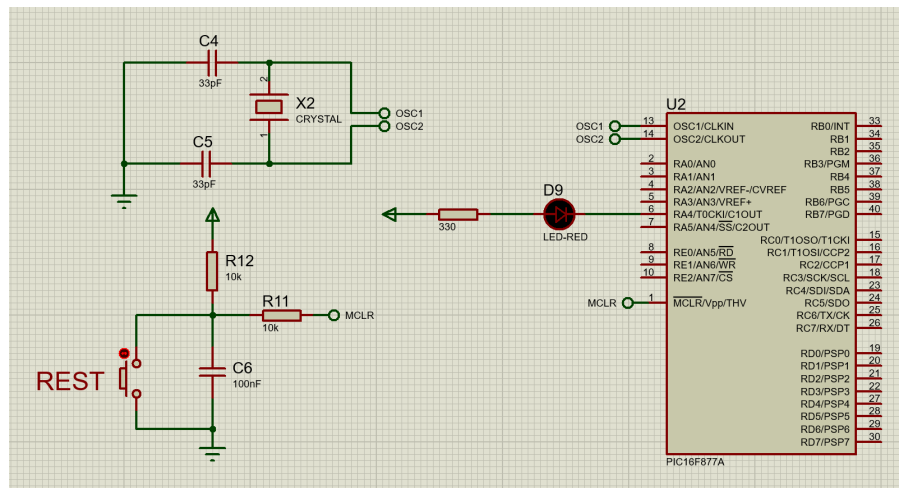


Figure 3: Conceptual active-low LED connection for RA4.

4.3 Example MikroC Code

A simple test program is:

```
void main() {

    ADCON1 = 0x06;           // Configure PORTA for digital use
    TRISA4_bit = 0;          // RA4 as output

    while(1) {

        RA4_bit = 0;         // LED ON
        Delay_ms(500);

        RA4_bit = 1;         // LED OFF
        Delay_ms(500);

    }

}
```

The exact ADCON1 configuration should always be selected according to the required analog channels in the complete project.

5 Part (d): ATmega328P versus PIC16F877A

The ATmega328P is an 8-bit AVR microcontroller, while the PIC16F877A is an 8-bit PIC microcontroller. Both devices are designed for embedded systems and provide digital I/O, timers, ADC, serial communication and non-volatile memory.

However, they differ significantly in memory capacity, number of pins, architecture, power-management features and peripheral capabilities.

The purpose of this comparison is to evaluate which microcontroller is more suitable for a particular embedded system rather than considering one device universally better than the other.

5.1 Memory Size Comparison

Memory capacity is one of the most important differences between the two microcontrollers.

Table 2: Memory comparison between PIC16F877A and ATmega328P

Memory Type	PIC16F877A	ATmega328P
Program Memory	8K $\times$ 14-bit words	32 KB Flash
SRAM	368 bytes	2 KB
EEPROM	256 bytes	1 KB

The ATmega328P has a clear advantage in memory capacity.

It provides 32 KB of Flash memory compared with the PIC16F877A's 8K instruction words. It also provides 2 KB of SRAM and 1 KB of EEPROM, which are considerably larger than the PIC16F877A's 368 bytes of RAM and 256 bytes of EEPROM.

This difference is particularly important in applications that require:

- Large software programs.
- Sensor and communication libraries.
- Large arrays and data buffers.
- Display libraries.
- Storage of configuration data.

Therefore, when memory capacity is a major system requirement, the **ATmega328P is the better choice**.

5.2 Pin Count and I/O Comparison

The number of available pins is another important factor in microcontroller selection.

Table 3: Pin and I/O comparison

Characteristic	PIC16F877A	ATmega328P
Common DIP Package	40 pins	28 pins
General-Purpose I/O	Up to 33	Up to 23
ADC Channels	Up to 8	6 in 28-pin package
ADC Resolution	10-bit	10-bit

The PIC16F877A has a significant advantage in the number of available physical I/O pins.

The PIC16F877A provides up to 33 general-purpose I/O pins, while the ATmega328P in its common 28-pin package provides up to 23 programmable I/O pins.

Therefore, a system containing many switches, LEDs, displays, sensors and actuators may benefit from the larger number of pins available on the PIC16F877A.

On the other hand, the smaller 28-pin package of the ATmega328P can be an advantage when PCB size and compactness are important.

Thus:

Many I/O connections → PIC16F877A

Smaller physical design → ATmega328P

5.3 CPU Architecture

Both microcontrollers use an 8-bit RISC-based architecture, but their instruction architectures are different.

The PIC16F877A uses the PIC architecture, while the ATmega328P uses the AVR architecture.

The ATmega328P has a more register-oriented AVR architecture with 32 general-purpose working registers. This can make many operations efficient and provides a convenient programming model.

The PIC16F877A uses a W (Working) register together with its file register architecture. Consequently, the two devices are both 8-bit microcontrollers, but their instruction sets, registers and programming models are not identical.

5.4 Timers and PWM

Both microcontrollers provide hardware timers that can be used for delays, event counting, periodic interrupts and waveform generation.

Table 4: Timer and PWM comparison

Feature	PIC16F877A	ATmega328P
Timer/Counter Modules	3	3
8-bit Timers	Timer0, Timer2	Timer0, Timer2
16-bit Timer	Timer1	Timer1
PWM Capability	2 CCP modules	Multiple PWM outputs

Both devices are therefore capable of controlling motors, generating PWM signals, producing accurate timing intervals and handling periodic events.

The exact number and configuration of PWM channels depend on the selected device package and peripheral configuration.

5.5 Analog-to-Digital Conversion

Both microcontrollers include a 10-bit ADC.

The PIC16F877A provides up to eight analog input channels, while the ATmega328P provides six analog input channels in its common 28-pin package.

Table 5: ADC comparison

Feature	PIC16F877A	ATmega328P
ADC Resolution	10-bit	10-bit
Analog Inputs	Up to 8	6

Therefore, the PIC16F877A is more suitable when a system needs a larger number of analog inputs without adding an external multiplexer or ADC.

5.6 Physical Size and Package

The PIC16F877A is commonly used in a 40-pin package, while the ATmega328P is commonly available in a 28-pin DIP package.

This difference affects the PCB design.

The PIC16F877A's larger package provides more pins, which is useful for systems with many external connections.

The ATmega328P's smaller package can be preferable when the application requires a compact PCB and does not need a large number of I/O pins.

5.7 Overall Comparison

The following table summarizes the most important characteristics.

Table 6: Overall comparison of PIC16F877A and ATmega328P

Characteristic	PIC16F877A	ATmega328P
Architecture	8-bit PIC architecture	8-bit AVR architecture
Program Memory	8K × 14-bit words	32 KB Flash
SRAM	368 bytes	2 KB
EEPROM	256 bytes	1 KB
Common Package	40-pin	28-pin
General I/O	Up to 33	Up to 23
ADC	Up to 8 channels, 10-bit	6 channels, 10-bit
Timers	Timer0, Timer1, Timer2	Timer0, Timer1, Timer2
PWM	2 CCP modules	Multiple PWM outputs
USART	Yes	Yes
SPI	Yes	Yes
I ² C	Yes	Yes
Low-Power Operation	SLEEP mode	Multiple power-saving modes
Memory Capacity	Lower	Higher
Number of I/O Pins	Higher	Lower
Compact Package	Less suitable	More suitable

5.8 Final Evaluation

The comparison shows that the ATmega328P is particularly attractive for systems that prioritize:

- Memory capacity.
- Compact physical implementation.
- Low-power operation.
- Software that requires more Flash and SRAM.

In contrast, the PIC16F877A remains a strong choice when the main requirement is a large number of physical I/O pins and analog input channels.

Consequently, the engineering decision should be based on the system specification rather than simply choosing one MCU as universally superior. ““